

Combining Surveys and Flow Logger Data to improve the accuracy of End-Use Segregation of Residential Water Consumption

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ABSTRACT

With rising water scarcity, managing water resources efficiently is becoming more important. Smart meters, provided they are of sufficient resolution, offer the potential to understand how people use water, to predict future demand, identify leakage and improve water efficiency. To successfully perform water end-use segregation, considerable manual input is currently needed. By utilising survey data, it may be possible to remove some of the human effort and find end-uses through pattern matching. In this paper, the use of survey and flow data shows a high distribution of repeated patterns in single and multiple-chain flow events for a particular brand of dishwasher.

Keywords: Water end-use segregation; Domestic water efficiency; Demand forecasting.

1 Background

Water scarcity is increasing and is predicted to affect 4 billion people by 2050 [1]. Water scarcity occurs when water resources are overused or managed inefficiently and is exacerbated during periods of drought. If not managed correctly, the future strain on water resources will increase due to climate change and growing populations. The ability to understand how water is used by households would allow resource managers to predict future demand, identify leakage and improve water efficiency. The potential adoption of smart metering for residential properties, gives an opportunity to employ end-use segregation to determine how water is being used [2].

1.1 Water End-use Segregation

End-use segregation of water consumption (otherwise known as micro-component analysis, or water disaggregation), breaks down the water consumption data into separate end uses, (e.g. shower, washing machine, toilet). There have been numerous studies into methods for disaggregating the flow, for example using decision trees, [3], [4], probabilistic functions [5] or Hidden Markov Models, Artificial Neural Networks and Dynamic Time Warping combined [6].

As described in [7], all these methods require significant human effort in the process, which is time consuming and costly. It is suggested in [7] that a more automated approach is needed. A non-dominated genetic sorting algorithm was employed for splitting the events. Additionally, clustering methods based on end-use characteristics (volume, duration, flow rate) were applied for the classification. However, unless there is a ground truth dataset to compare with, classification for any of the above-mentioned methods requires some human judgement. As an alternative in this paper, it is proposed to use domestic water consumption survey information to enhance the accuracy of classification, comparing patterns of use against information detailed in the survey and therefore removing part of the human judgement in the process.

1.2 Surveys

Previous usage of survey information for water consumption modelling has either been as a distinct water usage survey [8], [9], or for end-user profiling [10]. As far as the authors are aware, survey information has not been used in combination with water flow logger data to enhance the accuracy of the water end-use segregation, particularly on a dataset of this magnitude (detailed in dataset section). By using specific survey information on washing machine and dishwasher brands and models, it is proposed to increase the recognition of these appliance end-use types through repeated patterns within the flow trace. This paper will determine the feasibility of incorporating such data into a model, by looking at flow events between households that gave the same survey answers.

2 Methodology

Water consumption data was collected using Diehl Altair domestic water meters with Diehl Izar pulse units. These connect to a data logger, an Ashridge Engineering Textlog Flow 1, to transmit the data via cellular networks. The hardware setup is shown in Figure 1. The configuration has a volumetric resolution of 1 litre and sampling rate of 1 Hz, the data being transmitted in bulk every 24 hours, to save on battery life. The water meters have a wide distribution (about 1000 households) in the South West of the UK, Figure 2, incorporating both rural and urban locations.



Figure 1. Configuration for pulse readings

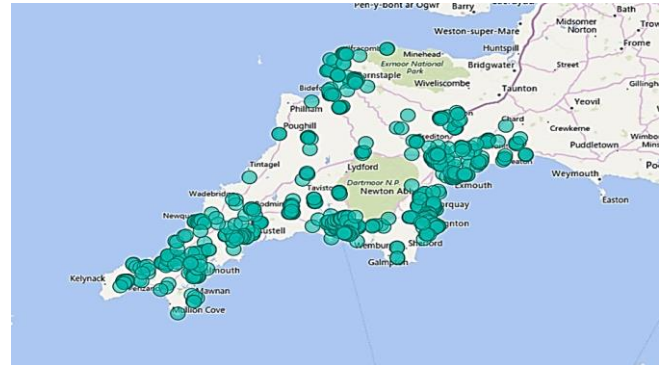


Figure 2. Meters with pulse loggers.

2.1 Pulse Flow Dataset

Table 1 shows an example of the pulse readings from a meter, with an identification column (MeterID), timestamp of the pulse (DTT) and the number of pulses received (No. Pulses), with each pulse representing 1 litre.

Table 1. Sample pulse data from a single household.

MeterID	DTT	No. Pulses
13834	03/06/2016 10:33:20	1
13834	03/06/2016 10:47:15	1
13834	03/06/2016 10:47:23	1
13834	03/06/2016 10:47:31	1

Using the pulse information, the flow rate is calculated by the number of pulses per time interval. Figure 3 shows an example of the flow trace, with a typical morning pattern of toilet and shower usage.

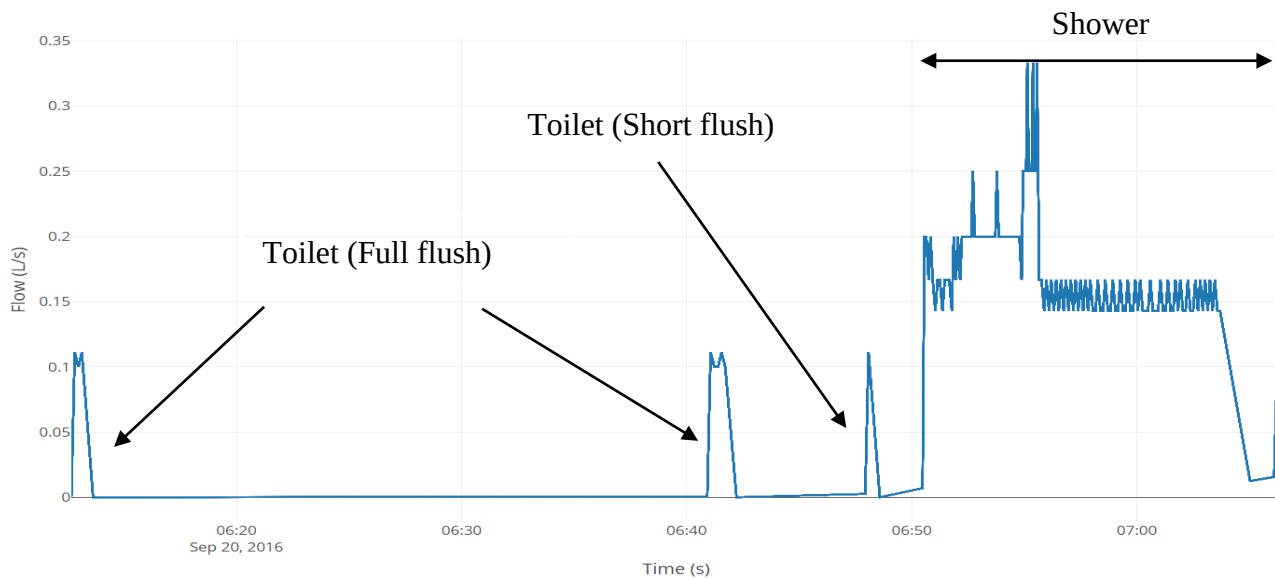


Figure 3. Flow trace for a single household.

2.2 Survey Dataset

South West Water conducts an annual survey of domestic water consumption for a specific set of households and has been doing so for a period of more than 40 years. The questions range from demographic (e.g. age and occupancy) and property information (e.g. house or flat, garden size) to appliances (e.g. dishwasher ownership) and water usage estimates (e.g. length of shower). This year's survey was sent out on 10th January 2018 to 1400 households and contained the following questions:

- No. of residents by age (16 and under, 17-29, 30-44, 45-64, 65+)
- No. of temporary residents by age. e.g. students (16 and under, 17-29, 30-44, 45-64, 65+)
- No. of daytime residents per week, e.g. retired, home workers (under 1 day, 1-3 days, 3+ days)
- Make and model of appliances (washing machine, dishwasher)
- No. of water saving devices (low flow showerhead, flow restrictors, low flush toilets, other)
- No. of showers per week by duration (Under 5 mins, 5-10, 11-15, 16-20, 20+)

Respondents were remunerated £7.50 for their participation with both online and paper versions of the survey. In total, 1039 households (74%) responded to the survey by April 2018 and of those who responded, 384 (37%) had active and functioning loggers, forming the set of data for analysis.

2.3 Variability and Limitations

Given that washing machines use a different amount of water based on the weight of the clothes inside, the focus of the analysis will look specifically at dishwashers, since these have a higher chance of regular events relative to washing machines. Of the 384 respondents with a logger, 176 either left the dishwasher question blank or stated they have no dishwasher. Out of the 208 households with a dishwasher, Brand A was found to be in use by 66 respondents (32%), therefore dishwashers of Brand A will be the target for analysis.

However, dishwashers of Brand A also have variability due to different programmes, 7-16 litres/use, Figure 4, or due to the level of water saving, between 3 to 6 litres/use, offered by technologies on some models.

	Auto 65°–75° auto	Auto 45°–65° auto	Eco 50° eco	Auto 35°–45° auto	Quick 45°	Prerinse
Duration: hs : mins	←	3:15 - 1:05	→	←	0:29 *	0:15
Energy consumption in kWh	←	1,35 - 0,60	→	←	0,70	0,05
Water consumption in litres	←	16 - 7	→	←	9	4

Figure 4. Overview of programmes taken from a summary of instructions.

The flow dataset from the 384 households was not filtered, but was divided into events, with a cut-off after a 30 second period without any pulses, giving a minimum flow of 2 litres/minute. This is lower than the minimum required flow in the dishwasher manual [11] at 10 litres/minute. The events did not consider multi-use activity, since at this stage the analysis is focused on determining if further investigation would be fruitful by looking for repeated patterns of activity between households with similar survey answers.

The volumetric resolution of the pulse also creates variability. Consider it as a tipping bucket, if an event uses 4.6 litres of water and there are already 0.4-0.99 litres in the bucket, the event will register 5 pulses, i.e. 5 litres. Whereas, if there are less than 0.4 litres already in the bucket, the event will register 4 pulses, i.e. 4 litres.

Despite the variability and limitations, given the large size of the dataset, it is hypothesised that there will be occasions when there is no water saving programme active and therefore the likelihood of repeated patterns will consequently increase.

2.4 Analytical Methods

Since the goal is to determine patterns across properties, the metric for determining repeated patterns will look at the number of properties with the same event characteristics (duration and volume). Once patterns are found, the events can be visualized to determine similarity.

The loggers' data was converted into flow events with characteristics of volume and duration, for example, 1 litre in 15 seconds (1x15), 1 litre in 16 seconds (1x16), 4 litres in 30 seconds (4x30) etc. This was achieved by utilising Python 3 code (with Pandas, Numpy). The result was imported into Excel and a pivot table used to find the number of properties with same event.

3 Analysis and Discussion

3.1 Individual Events

Figure 5 and 6 show a subset of events for the households with and without Brand A dishwashers, respectively. A closer analysis of Figure 5 shows a high distribution of events, between 30-60%, for the Brand A dishwasher ownership households. On the other hand, Figure 6 shows a much lower distribution, between 7-18%, for the households without Brand A dishwashers. This demonstrates that these events are more frequent for the households fitted with Brand A dishwasher.

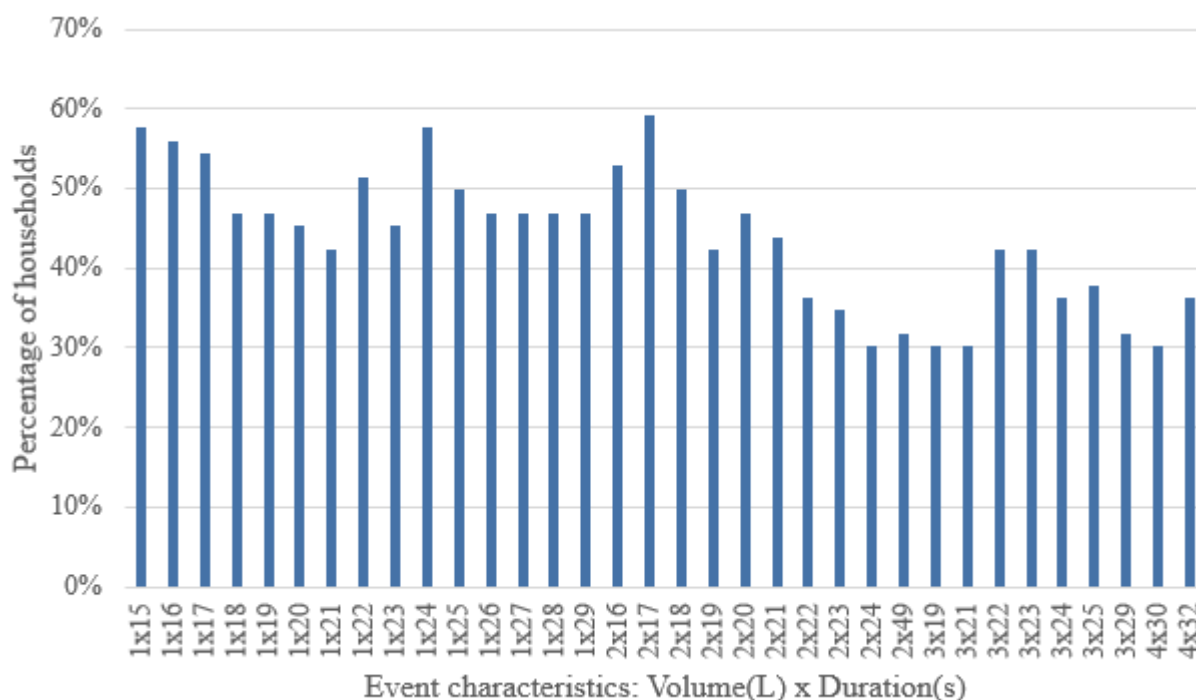


Figure 5. Percentage of households owning Brand A dishwashers with events of the same volume and duration and a distribution higher than 29%

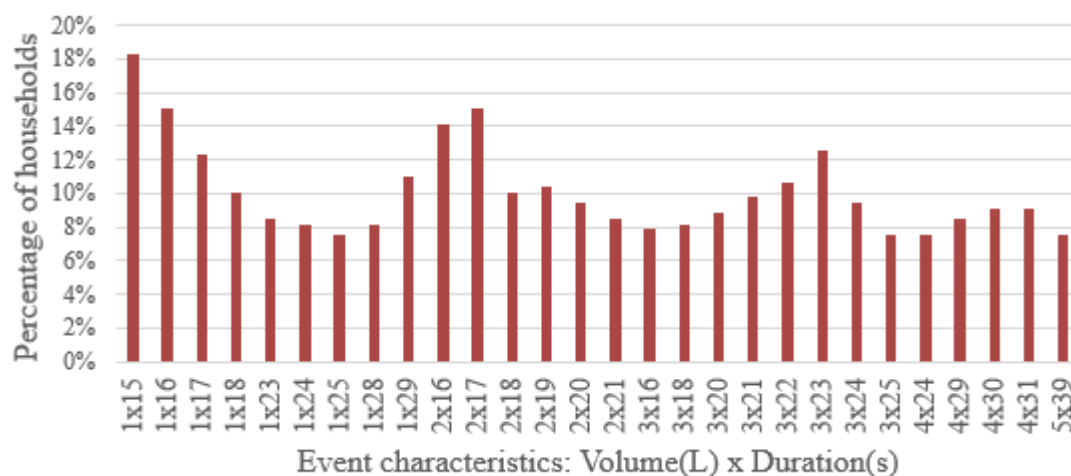


Figure 6. Percentage of households without Brand A dishwashers with events of the same volume and duration and a distribution higher than 7%

3.2 Chained Events

Section 3.1 provides a snap shot of individual events analysis, in this section, chained events (i.e. an event followed by another event) are discussed. Given that dishwashers have multiple phases, a frequency analysis was evaluated of chained events, e.g. “2x47,2x48”, i.e. 2 litres in 47 seconds event followed by 2 litres in 48. The chains can be of length 2 or length 3.

The 2-chain results shown in Figures 7 and 8 show the percentage of events for households with and without Brand A dishwashers respectively. A comparison of the two figures indicates a higher frequency of 2 chain events for households with the Brand A dishwasher. A similar observation can be made for the 3-chain results shown in Figures 9 and 10.

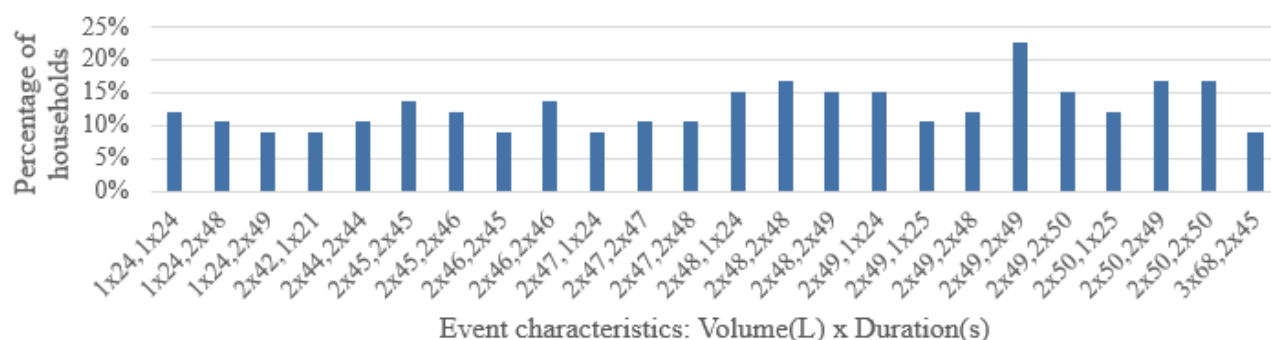


Figure 7. Brand A dishwasher households with 2-chain events and distribution higher than 7%

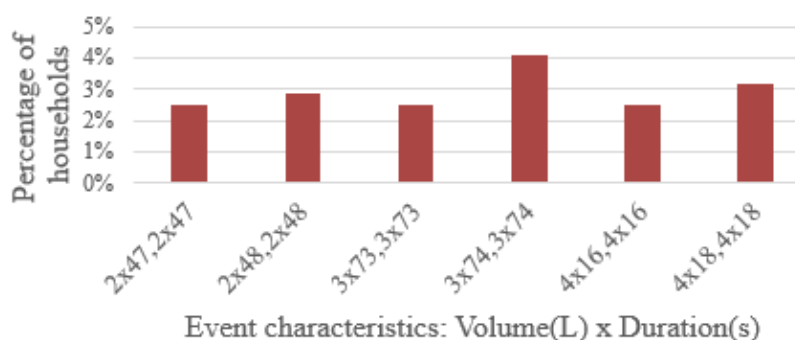


Figure 8. Non-Brand A dishwasher households with 2-chain events and distribution higher than 2%

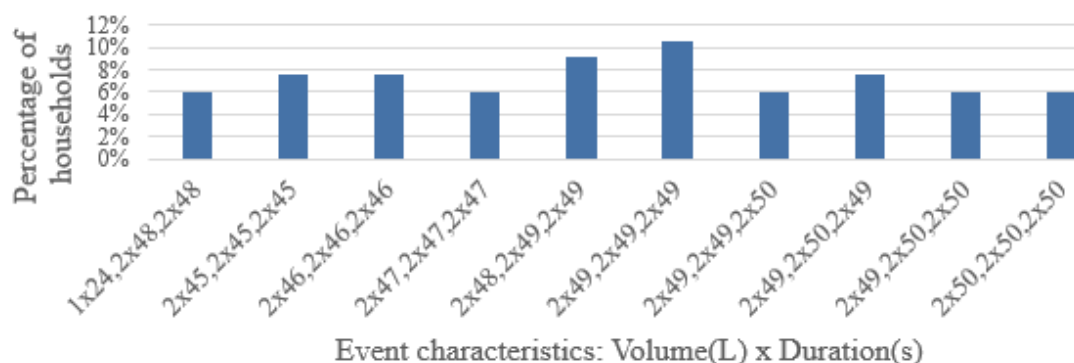


Figure 9. Brand A dishwasher households with 3-chain events distribution higher than 5%

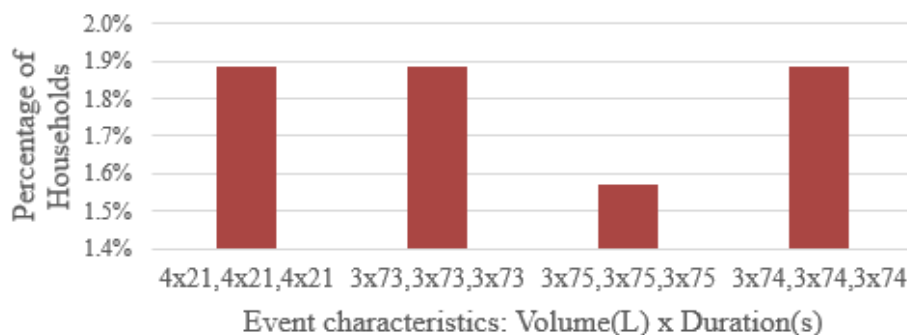


Figure 10. Non-brand A dishwasher households with 3-chain events distribution higher than 1%

3.3 Flow Trace Comparison

To examine the similarity of Brand A events, two households with the highest frequency of the 3-chain event, 2x49,2x49,2x49, were selected for flow comparison, shown in Figure 11, Figure 12.

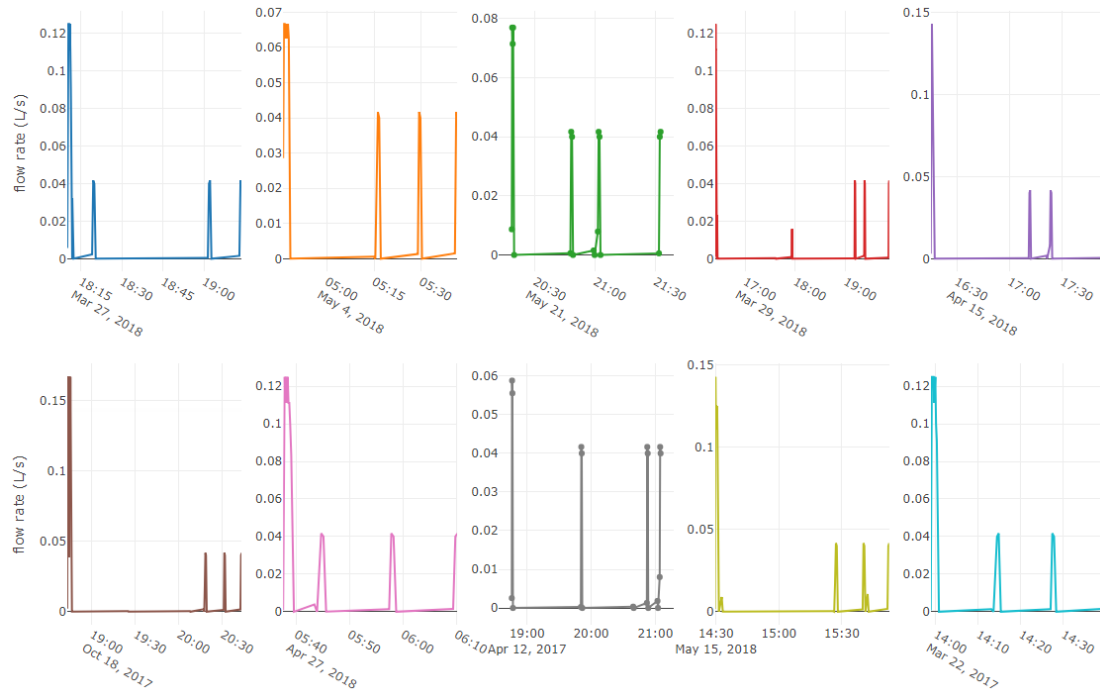


Figure 11. Flow traces (L/s) for property 19853, 2x49,2x49,2x49 events.

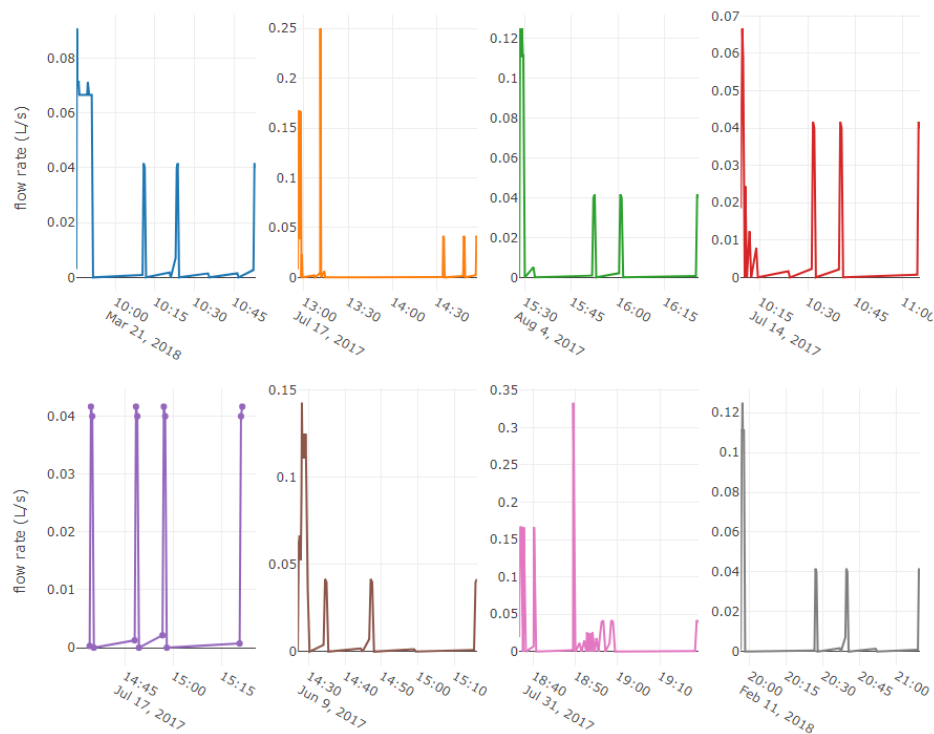


Figure 12. Flow traces (L/s) for property 20837, 2x49,2x49,2x49 events.

The flow trace demonstrates some similarities between events, with a single higher peak (variable in size) being followed by three peaks of around 0.04 l/s, generally having a total time duration of 1 hour and 15 minutes. Those with a different time period may have had other water events occurring at the same time or are not related to the dishwasher.

3.4 Future Work

Further characteristics e.g. peak flow, minimum flow and total duration may be useful in examining linked events. By examining the time phases between each single event, it is possible to determine sets of cycles for different dishwasher brands. This allows identification of patterns in situations where there is no distinct dishwasher use, i.e. if other events are occurring during the period of the dishwasher use. After further validation, this method could also be used to identify other appliances.

4 Conclusions

End-Use segregation of water is a complex problem and if the goal is to fully automate the process, then methods that increase certainty in identification will be vital. The use of survey information to identify dishwasher patterns warrants further investigation and with detailed analytics, may find appliances with almost absolute certainty. A large dataset is required for analysis and if limitations in resolution can be overcome, then this approach can potentially lead towards more reliable appliance identification and therefore support decision making for water efficiency programmes.

5 References

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